Tele Project Deployment and Setup Report

Introduction

This report provides detailed steps for setting up and deploying the Tele project, which is hosted on GitHub and deployed on the Google Cloud Platform (GCP). It covers both local setup and server deployment.

Project Overview

- Project Name: Tele
- Repository: *https://github.com/maniksejwal/tele*
- Deployment Platform: Google Cloud Platform (GCP)
- Technologies Used: React.js, Node.js, Express.js, MongoDB, PM2, Nginx

Local Setup

Follow these steps to set up the project locally for development and testing.

Clone the Repository

- Clone the repository:
  git clone https://github.com/maniksejwal/tele

- Navigate to the project directory:
  cd tele

Frontend Setup

- Navigate to the frontend folder:
  cd frontend

- Install dependencies:
  yarn install

- Start the frontend:
  yarn start

Backend Setup

- Navigate to the backend folder:
  cd backend

- Install dependencies:
  yarn install

- Start the backend server:
  yarn start

Deployment on GCP

Prerequisites

- Ensure the GCP instance is set up with SSH access.
- Install Nginx, PM2, and Node.js on the server.
- The project should be stored in a directory on the instance (e.g., /var/www/tele).

Deploying the Frontend

- Check if the frontend has a build folder:
  ls frontend/build
  If not, build it:
  yarn build
- Create a compressed archive of the build folder:
  tar czf build.tar.gz build
- Transfer the build archive to the GCP server:
  scp build.tar.gz user@server-ip:/path/to/frontend
- SSH into the GCP server:
  ssh user@server-ip
- Navigate to the frontend directory on the server:
  cd /path/to/frontend
- Remove the existing build folder:
  rm -r build build.tar.gz
- Extract the new build archive:
  tar xzf build.tar.gz
- Move the build folder to Nginx's root directory:
  mv build /var/www/tele/ or (rm, then cp)
- Restart Nginx:
  sudo systemctl restart nginx

Deploying the Backend

- Transfer updated backend files (e.g., questions.js):
  scp backend/questions.js user@server-ip:/path/to/backend/
- SSH into the server and navigate to the backend folder:
  ssh user@server-ip
- *cd /path/to/backend*
- Restart the backend service using PM2:
  pm2 restart 0